Digital Twins for Predictive Maintenance in Automotive Aftersales

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***Abstract*—TODO: Abstract here.**

***Index Terms*—Digital Twin, Predictive Maintenance, Industry 4.0, Machine Learning, IoT, Condition Monitoring**

I. Introduction

Today's vehicles are connected by default and continuously transmit operating data [1], [2], so the information needed to judge their technical condition largely exists. Yet aftersales remains reactive: even well-connected vehicles are serviced on fixed time or mileage intervals rather than on the actual condition of their components. Wear parts—brake pads and discs, tyres, ignition parts, or wipers—are replaced only once a fault becomes visible, so customers run them to failure and the unplanned repair often goes to a cheaper third-party garage. For the dealership, which owns the customer relationship, this means losing both the repair and one of only a few yearly customer contacts—each also a chance to advise, retain, and sell [3], [4].

Digital-twin-based predictive maintenance (PdM) can turn this data into foresight. A digital twin (DT) maintains a synchronised virtual representation of the vehicle [5], against which machine-learning models forecast the remaining useful life of components and recommend replacement before failure—keeping the customer at the dealership and enabling transparent, targeted upselling. While the building blocks of DT-based PdM are well studied in industrial and aerospace settings [6], [7], their integration into automotive aftersales remains underexplored.

This paper therefore asks: *what effect does the proactive use of digital twins and predictive maintenance have on the automotive aftersales domain?* Focusing on the German market and working with explicitly stated assumptions, we (i) clarify the interplay of DT and PdM in aftersales, (ii) compare the predictive target state with the interval-based current state to reveal its potentials, and (iii) derive recommendations for the dealership. The paper is structured along the IEEE format: theoretical background, methodology, gap analysis, discussion, and conclusion.

II. Related Works & Theoretical Background

A. Digital Twins

The concept of the digital twin was first introduced by Grieves [5] in a white paper on manufacturing excellence, where it described a virtual replica of a physical product maintained in synchrony with its real-world counterpart throughout the entire asset lifecycle. Originally conceived as a tool for improving production quality in industrial manufacturing, the idea drew on advances in simulation, sensor technology, and computational modelling to propose that a persistent virtual representation of a physical object could serve as the basis for monitoring, analysis, and decision-making. Over the subsequent decade, the concept migrated from its manufacturing origins into aerospace, energy, and infrastructure domains, before emerging as a foundational concept in Industry 4.0 discourse more broadly [6], [20].

In the automotive domain, this general architecture maps directly onto the connected vehicle. The physical entity is the vehicle itself, equipped with an array of on-board sensors and an On-Board Diagnostic (OBD) interface; the virtual representation is a cloud-hosted model that mirrors component states, degradation histories, and operating conditions; and the synchronising link is the over-the-air (OTA) telematics connection that transmits telemetry in near real time [1], [2]. Lopes et al. [1] situate this configuration within a broader software ecosystem in which the digital twin acts as the authoritative source of vehicle state for all downstream services. Architecturally, automotive digital twins are typically realised as modular, microservice-based platforms: a data ingestion layer normalises heterogeneous sensor streams—such as vibration, temperature, voltage, and Controller Area Network (CAN)-bus signals—before forwarding them to a model layer that maintains the current state of each monitored component [8], [10]. The outputs of this stack are then exposed to external consumers such as workshop management systems, Original Equipment Manufacturer (OEM) backends, and customer-facing notification services [1], [7], forming the basis for different predictive maintenance strategies.

B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [7]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [7], [8]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [8]; and PdM extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until end-of-life is reached [7], [9].

PdM relies on three data layers: on-board sensors delivering high-frequency signals such as vibration, current, voltage, and cell temperature [8]–[10]; the OBD interface providing standardised diagnostic codes [2], [11]; and OTA telematics enabling continuous low-latency exchange with the backend [1], [2], [10]. Together, these layers transform the vehicle from a periodically inspected asset into a continuously observed system, a shift first documented in 2009 by Zhang et al. [11].

PdM at this scale is not a stand-alone technique but a service layer built on a DT of the vehicle. As Lopes et al. [1] and Hadraoui et al. [8] describe, the DT supplies the synchronised virtual representation against which sensor streams are interpreted, anomalies detected, and remaining-useful-life estimates computed. Without a DT, telemetry must be processed in isolation; without PdM, the DT remains a passive monitor. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [7], [10].

C. Machine Learning in PdM

Three problem classes structure machine-learning (ML)-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks on time-frequency representations of vibration signals [7], [9]. Regression targets Remaining Useful Life (RUL) estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [7], [10]. Anomaly detection operates without explicit labels, learning nominal behaviour and flagging deviations as failure precursors [7], [10]. AbouGrad et al. report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they emphasize that these figures depend strongly on dataset quality and operating conditions.

A recurring constraint is data: AbouGrad et al. and Hadraoui et al. note that preprocessing consumes 70–90% of project time, genuine failure data is structurally scarce, and public benchmarks representative of in-service behaviour are largely absent. Model-based and hybrid approaches that embed physical degradation laws into the learning pipeline are therefore preferred over purely data-driven ones, as they extrapolate more reliably under data scarcity [6], [8], [9]. Beyond accuracy, deployment in safety-relevant contexts raises the question of explainability: deep architectures achieve high accuracy but operate as opaque models whose decisions cannot readily be justified to engineers, customers, or regulators [7]. Explainable AI (XAI) techniques such as feature attribution and attention-based saliency are increasingly integrated into PdM pipelines to expose the reasoning behind a prediction, marking the conceptual bridge from smart to wise information systems [7].

D. Automotive Aftersales

Aftersales encompasses post-purchase activities that help customers access supplier services and resolve product issues through continuous maintenance and upkeep [12]. As a quality-focused product support activity, it creates a sustainable competitive advantage across industries, with automotive being a prime example [13], [14]. In the automotive context, it functions as a dedicated business module bundling maintenance, repair, and warranty offerings to maximise product use throughout the asset lifecycle while meeting financial and customer-performance targets [4], [15]. Building post-sales relationships strengthens customer loyalty, yields actionable insights, and creates upselling and cross-selling opportunities [16], making the automotive aftersales market increasingly attractive as demand for higher-quality service expands [4].

Historically, customers of maintenance-free vehicles lose interest once the warranty expires. Within the automotive industry, aftersales services represent a particularly significant source of revenue, with profit margins considerably exceeding those generated through vehicle sales itself [17]. From an economic standpoint, the aftersales service market has been identified as substantially larger than the primary product sales market across various industries [18]–[20]. To retain these customers, better information systems and service processes are essential [12]. The growing proactive recognition that anticipating issues before they arise improves performance, efficiency, and long-term operational excellence [21].

Proactive aftersales improves customer satisfaction and retention by resolving issues before customers encounter them, enabling more productive interactions and innovation [3]. This requires tailoring services to customer needs through tight integration of operational and informational data [12]. ML and DT-based approaches enable predictive interventions and better asset lifecycle optimisation, delivering cost reduction, improved customer satisfaction, and extended asset lifetimes [22].

E. Synthesis and Research Gap

The preceding subsections establish three building blocks: the DT as a synchronised vehicle model, PdM as a prognostic analytics layer built on top of it, and aftersales as the customer-facing business context in which service decisions are communicated and acted upon. Together, they form a potential end-to-end system in which the DT continuously feeds component-state data to ML-based PdM models, whose forecasts are surfaced through aftersales processes to enable proactive service appointments and retain customers at the dealership.

While the referenced contributions establish the technical foundations of DT-based PdM, existing work concentrates either on algorithmic aspects in isolation [7], [9] or on industrial applications [6]. The integration of DT-based PdM into automotive aftersales business models, particularly regarding its implications for customer interaction, remains underexplored. The present work addresses this gap.

III. Methodology

This paper aims to analyse recent academic studies on applying DT and PdM in the automotive industry. A Literature Review (LR) was performed to achieve this objective. A central element of this method is to identify, assess and interpret current research in a structured, transparent as well as replicable way [23]. Thus, adopting the methodological approach of a LR helps minimise bias arising from selective literature selection, while simultaneously outlining the current state of the art on DT and PdM in the automotive industry. Furthermore, it serves to pinpoint existing research gaps as well as future research needs within this respective field [24].

Subsequently, a gap analysis will be performed as a further analytical step within this paper. The central objective of this analysis is to identify and assess the differences between the current state of aftersales in the automotive industry and its potential future development, with particular regard to the combined application of DT and PdM. By systematically comparing existing practices with emerging research findings and technological possibilities, the gap analysis aims to highlight areas where significant advancement is yet to be achieved. In doing so, it serves as a basis for deriving concrete implications and future research needs in this rapidly evolving field [25].

IV. Gap Analysis

A. Lifecycle Comparison

B. Assumption Framework

C. Potential Assessment

D. Limitations

V. Discussion and Recommendations

- Interpretation der Ergebnisse
- Implikationen für Forschung und Praxis

VI. Conclusion & Future Work

- Zusammenfassung der wichtigsten Erkenntnisse
- Beitrag der Arbeit
- Ausblick auf weitere Forschung

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Erklärung zur Nutzung von KI-Tools

TODO: KI-Erklärung hier einfügen (gemäß DHBW-Vorgabe).